



GOOGLE CLOUD · BUILD WITH AI: CODE FOR COMMUNITIES 2026 · TRACK 4

Kisan Alert

Satellite-grounded irrigation, crop, fertilizer and plant-health advice for small and marginal farmers — in Telugu, Hindi and English, over web, voice and SMS.

Live · rajneeshrehsaan-kisan-alert.hf.space

Code · github.com/Rajneesh180/kisan-alert

THE PROBLEM

Frontline farm advisory still runs on paper registers and phone calls.

A smallholder needs three answers in time to matter: **irrigate or wait? what to sow? how much fertilizer?** Today those answers arrive late, in the wrong language, or not at all.

Timing decides the harvest

A dry spell or an irrigation missed by a week can cost a season — but weather and satellite signals reach the farmer slowly, if ever.

Language shuts people out

Most advisory tools are English-first and text-heavy — unusable for a low-literacy Telugu- or Hindi-speaking farmer on a basic phone.

Data sits in silos

Sentinel-2, IMD/ERA5 climate, SoilGrids and groundwater records exist — but never arrive together as one plain-language "what to do today".

The gap is not more data. It is turning scattered public data into a single act a farmer can take *today*.

THE SOLUTION

One glance: what to do, on this plot, today.

Pick a plot and the top of the screen answers first — an action-first verdict, then the evidence behind it.

Action-first verdict

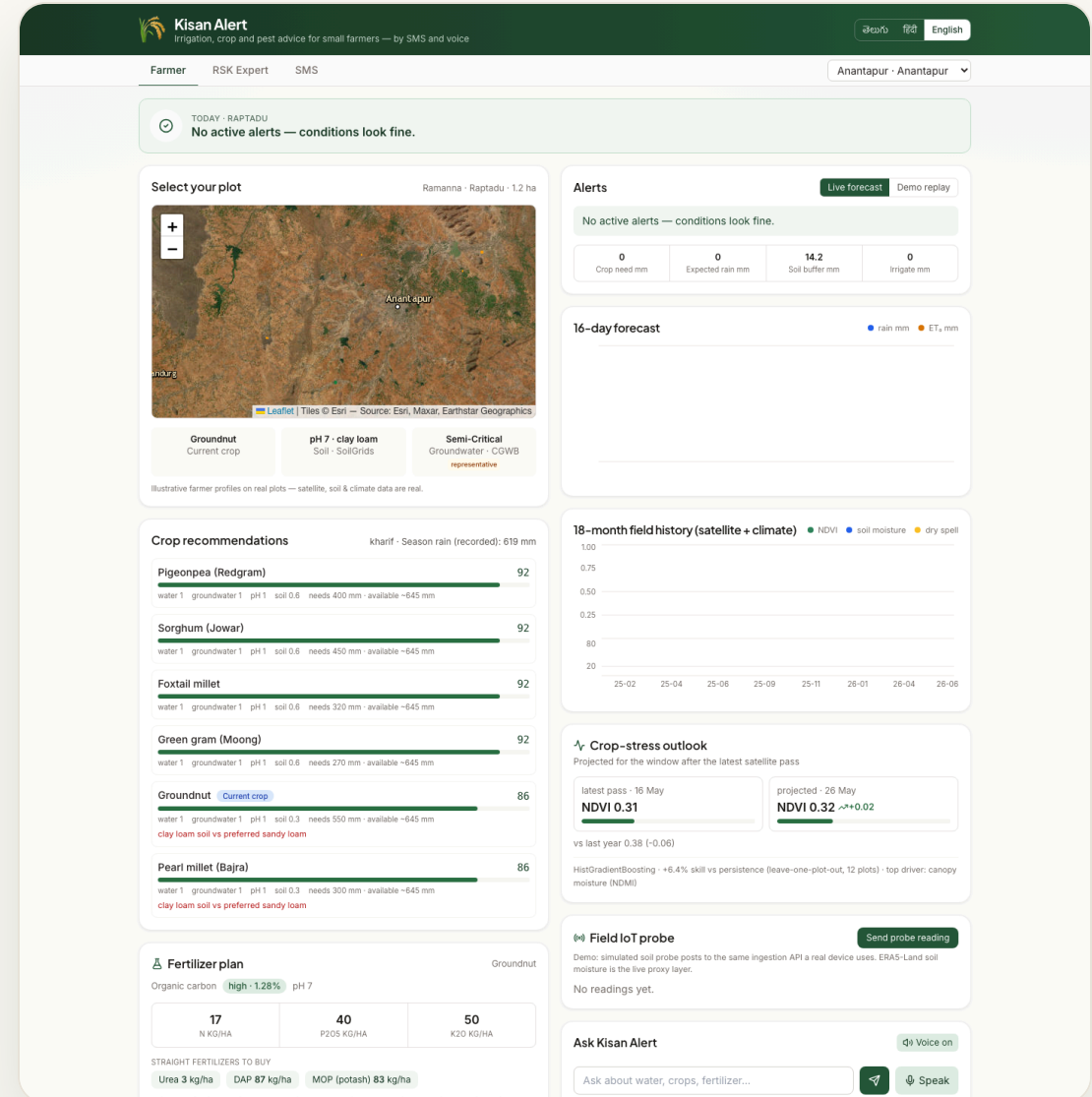
"Irrigation due — ~44 mm this week", not a wall of charts.

Every number is grounded

Water balance, dry-spell, crop-stress and fertilizer come from deterministic agronomy engines on real data.

Trilingual, three channels

Telugu / Hindi / English — over web, voice and SMS.



Live farmer dashboard — real Esri/Sentinel imagery, crop scores, NPK plan, crop-stress outlook

The model that must be right computes the numbers. Gemini only speaks them.

Deterministic engines

FAO-56 water balance · dry-spell detection · NDVI seasonal-anomaly stress · crop-suitability · soil-test NPK. Every figure a farmer acts on is computed here — **Gemini is never allowed to invent a measurement.**

Machine learning

A gradient-boosted 10-day crop-stress forecast on 9 agronomic features. Benchmarked honestly: **+6.4% skill over a persistence baseline** under leave-one-plot-out CV. Top driver: canopy moisture (NDMI).

Gemini — language layer

Grounded advisory, Indic speech-to-text, multimodal crop-photo diagnosis, native Telugu/Hindi text-to-speech, and RAG over ICAR/TNAU practices with text-embedding-004.

Grounded, live example. Asked "when should I irrigate my paddy this week?", the deployed assistant answered: *"our records show your current crop is cotton, not paddy... water need 0 mm, no irrigation needed this week."* — it corrected the crop from plot data and used the computed water balance, instead of guessing.

Offline-first: with no AI key or on any Gemini failure, every feature falls back to its deterministic result.

Real public data → engines → farmer's language.

1 · Ingest (offline pipeline)

Sentinel-2 NDVI/NDMI via Earth Engine ·
Open-Meteo ERA5 rain/ET₀/soil-moisture ·
ISRIC SoilGrids · CGWB groundwater. 18
months per plot, committed as artifacts.

2 · Reason (FastAPI + SQLite)

Deterministic agronomy engines + the
gradient-boosted crop-stress model,
trained at boot from the committed data —
the served model never drifts.

3 · Deliver (React + Gemini)

Trilingual web app, voice, and
SMS/WhatsApp. Gemini phrases the
numbers; a human RSK expert catches
low-confidence photo cases.

Grounded in real signals, not demo stubs

12 plots across two constituencies — Anantapur and the sponsoring
MP's Narasaraopet. Farmer profiles are illustrative; the satellite, soil
and climate data under them is real.

Google Cloud, on free tiers

Gemini API (advisory, photo, TTS, RAG embeddings) and Earth
Engine (Sentinel-2) — both on free / noncommercial tiers. No billing
dependency.

Built for the farmer's phone, not the analyst's laptop.

Trilingual by default

Telugu, Hindi, English — every screen, alert and spoken answer.

Voice for low literacy

Ask by speaking; answers are read aloud in the farmer's language.

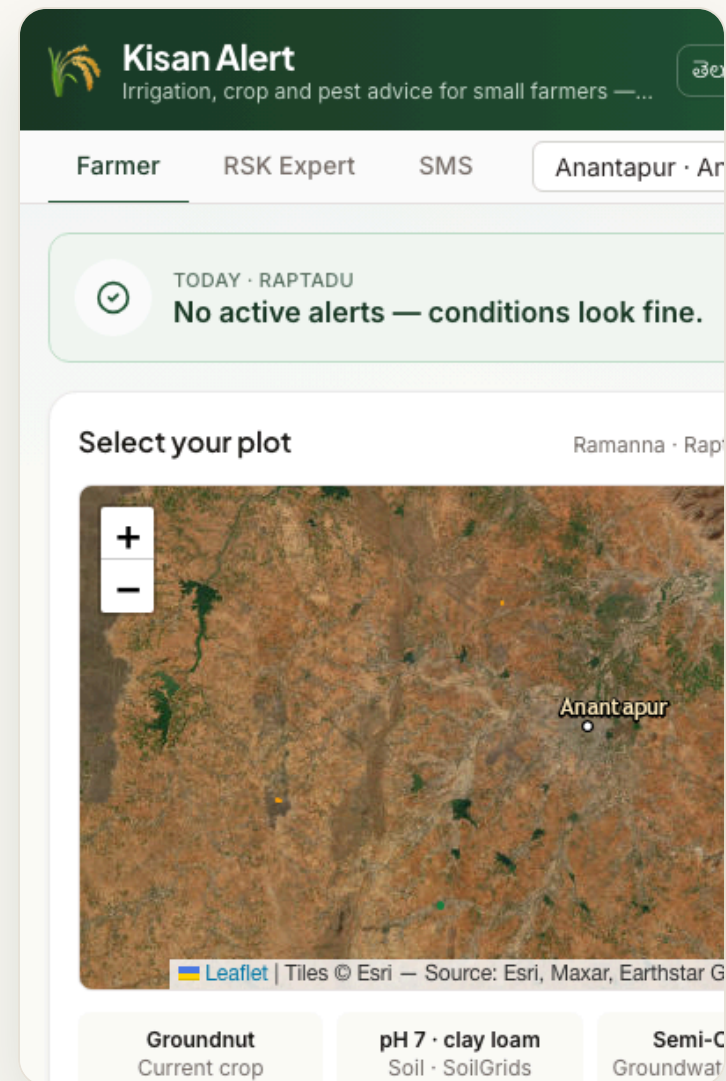
SMS / WhatsApp reach

Text a keyword, get the alert back — works on a basic phone, low connectivity.

Human safety net

Unsure photo diagnoses escalate to a Rythu Seva Kendra expert, not a guess.

WCAG-AA contrast, mobile-first layout, and a deterministic core so the tool still works when the network — or the AI — does not.



The same advisory on a phone

Running in a browser today; running in a constituency in weeks.

It already runs

One Docker image builds the frontend and serves the API. Deployed live on Hugging Face Spaces — open the link and use it, no setup.

Short, honest path to production

SQLite → Cloud SQL / BigQuery · manual pipeline → Cloud Scheduler GEE + Open-Meteo pulls · in-process limits → Memorystore · SMS simulator → the wired Twilio webhook.

12→∞

plots today; the pipeline and schema are per-plot, per-constituency from day one.

~1s

model train at boot; the whole app is one stateless container.

₹0

infra cost to demo — free tiers only, no billing to enable.

A pilot in Narasaraopet needs plot boundaries and a Twilio number — the rest is already built.

Every smallholder with a phone gets an agronomist's answer in their own language.

Who benefits

Small and marginal farmers first — the majority who are priced out of private agri-advisory.

Why it can scale

Public, keyless data + free-tier AI + a per-plot schema means a new mandal is data, not new code.

Ready for pilot

Built on Narasaraopet's own plots — ready to test with the sponsoring MP's office.

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